PROGRAM 9: Local RAG System Using Engineering Documents

AIM

To implement a Retrieval-Augmented Generation system using engineering documents, a vector database and a pretrained language model.

ALGORITHM

1. Import the required libraries.
2. Load engineering documents.
3. Generate document embeddings.
4. Store the embeddings in ChromaDB.
5. Accept a technical question.
6. Retrieve relevant documents.
7. Generate an answer using a pretrained language model.
8. Display the retrieved information and answer.

CODE

```
import chromadb
from sentence_transformers import SentenceTransformer
from transformers import pipeline


# Load embedding model
embedding_model = SentenceTransformer(
    "all-MiniLM-L6-v2"
)


# Load language model
llm = pipeline(
    "text2text-generation",
    model="google/flan-t5-small"
)


# Create vector database
client = chromadb.Client()
```

```
collection = client.create_collection(
    name="engineering_docs"
)

# Engineering documents
documents = [
    "CPU is the central processing unit of a computer.",
    "RAM is temporary memory used by running programs.",
    "An operating system manages computer hardware and software.",
    "A database stores and organizes information.",
    "Computer networks allow devices to communicate."
]

# Generate embeddings
embeddings = embedding_model.encode(
    documents
).tolist()

# Store documents
collection.add(
    documents=documents,
    embeddings=embeddings,
    ids=["1", "2", "3", "4", "5"]
)

# User question
question = input("Enter your technical question: ")

# Retrieve documents
```

```
query_embedding = embedding_model.encode(
    [question]
).tolist()

results = collection.query(
    query_embeddings=query_embedding,
    n_results=2
)

context = "\n".join(results["documents"][0])

# Generate answer
prompt = """
Answer the question using the context.

Context:
""" + context + """

Question:
""" + question

answer = llm(
    prompt,
    max_new_tokens=100
)

print("\nRetrieved Information:")
print(context)

print("\nGenerated Answer:")
```

print(answer[0]["generated_text"])

INPUT

Enter your technical question:

What is the function of RAM?

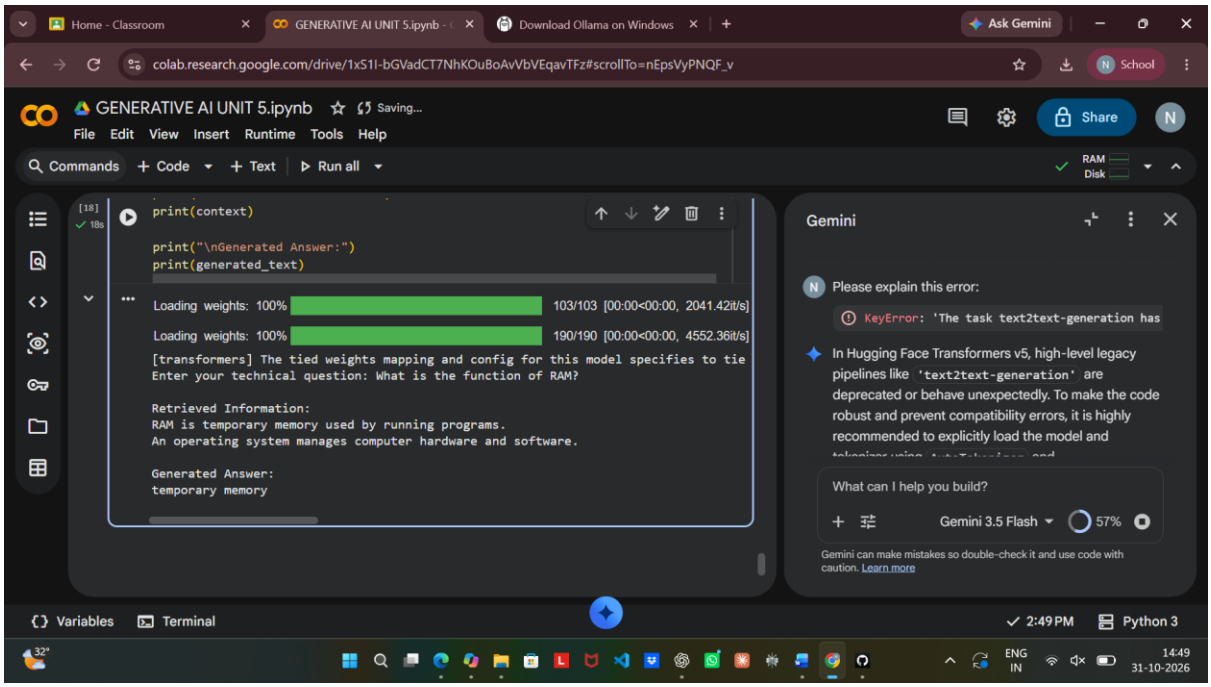
OUTPUT

Retrieved information:

RAM is temporary memory used by running programs.

Generated answer:

RAM is temporary memory used by running programs.



RESULT

Thus, a RAG system was implemented using engineering documents, ChromaDB, Sentence Transformers and a pretrained language model.